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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	Ⅰ-Ⅲ Semester(2025-26)
Faculty Name	Dr Jayaram Boga
Student Name	Ramannagari Bhanu Priya
Roll Number	2520090171
Branch	CSIT
Course Outcome	3

Case Study Topic: CampusFlow – University Campus Resource Management System using BFS

Work Done Summary:

Implemented Breadth First Search (BFS) in Java for campus infrastructure monitoring and resource traversal. Modeled classrooms, laboratories, faculty rooms, and campus facilities as graph nodes connected through edges. BFS traversal was used to monitor campus resources level by level and analyze facility connectivity efficiently.

Implemented classroom allocation monitoring, laboratory connectivity analysis, faculty scheduling traversal, student service request tracking, and campus route analysis using graph traversal techniques. Demonstrated BFS traversal for auditing connected campus facilities and analyzing campus resource flow efficiently. Compared BFS traversal with DFS — BFS is more suitable for level-wise monitoring and shortest-path based campus analysis.

Implementation Details :

1. Created Graph class using adjacency list representation.

2. addEdge() connects classrooms, laboratories, and campus facilities.

3. BFS(startNode) uses Queue data structure for traversal.

4. Visited array prevents duplicate traversal of nodes.

5. BFS traversal monitors campus facilities level by level.

6. Campus connectivity analysis performed using graph traversal techniques.

7. The main method created a campus graph and demonstrated BFS traversal.

8. BFS ensures efficient campus monitoring and infrastructure management operations.

Result : Screenshots/Output:

```
CampusFlow - University Campus Resource Management System
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Campus Connections:
Admin Block <-> Classroom Block
Admin Block <-> Computer Lab
Classroom Block <-> Library
Classroom Block <-> Faculty Room
Computer Lab <-> Canteen
Library <-> Auditorium

BFS Traversal:
Visited Facility : Admin Block
Visited Facility : Classroom Block
Visited Facility : Computer Lab
Visited Facility : Library
Visited Facility : Faculty Room
Visited Facility : Canteen
Visited Facility : Auditorium

Campus Monitoring:
Checking campus facility connectivity...
All campus facilities are connected.
Campus resource monitoring completed successfully.
```

Time Complexity:

- BFS Traversal: O(V + E)
- Graph Construction: O(E)

BFS is efficient for campus connectivity analysis, infrastructure monitoring, and campus resource management systems.

GitHub Link :https://github.com/ramannagaribhanupriya/DSA-2-PROJECT

Student Signature	**Faculty Signature**